Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Albert Mendes Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
1100 Ruble RD	Crows Landing	Stanislaus	95313

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1926

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X058-X003-X014-XXXX

B. OPERATORS

Mendes, Albert

Operator name: Mendes, Albert

Telephone no.: (209) 632-0350 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1100 Ruble RD	Crows Landing	CA	95313

This operator is responsible for paying permit fees.

C. OWNERS

Mendes, Albert

Legal owner name: Mendes, Albert

Telephone no.: (209) 632-0350 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
1100 Ruble RD	Crows Landing	CA	95313

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	300	200	0	300	0
Number under roof	1,400	0	0	0	0	0
Maximum number	1,400	300	200	0	300	0
Average number	1,400	300	200	0	300	0
Avg live weight (lbs)	1,400	1,400	900	0		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 42,969.68 *tons per reporting period*

Total nitrogen from manure: 554,295.65 *lbs per reporting period*

After ammonia losses (30% loss applied): 388,006.96 *lbs per reporting period*

Total phosphorus from manure: 90,222.01 *lbs per reporting period*

Total potassium from manure: 266,273.32 *lbs per reporting period*

Total salt from manure: 728,175.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 41,382,683 *gallons*

Total nitrogen generated: 145,974.69 *lbs*

Total phosphorus generated: 13,823.32 *lbs*

Total potassium generated: 140,528.58 *lbs*

Total salt generated: 1,899,306.15 *lbs*

	41,382,683	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	41,382,683	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
East Irrigation Well	Ground water
TID Lateral 4	Ground water
West Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/20/2023	Solid commercial fertilizer *NH3*	11.00 *ton*	Dry-weight	0.1	82.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	18,021.96	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	18,021.96	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	Corral solids	432.00 *ton*	Dry-weight	59.6		17,200.00	7,320.00	45,000.00		42.10
06/27/2023	Corral solids	1,070.00 *ton*	Dry-weight	59.6		17,200.00	7,320.00	45,000.00		42.10

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	20,874.20	8,883.67	54,612.72	510,932.34
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	20,874.20	8,883.67	54,612.72	510,932.34

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Bernies 10	10	10	2	process wastewater	0058-0003-0016-0000
Big 68	68	68	2	process wastewater	X058-X003-X024-XXXX X058-X003-X033-XXXX
Little 10	10	10	0	none	X058-X003-X033-XXXX
Long 35	35	35	2	process wastewater	0058-0003-0024-0000
Long 57	57	57	2	process wastewater	0058-0003-0021-0000 0058-0003-0025-0000
McWharters 40	40	40	2	process wastewater	0058-0022-0010-0000
Ruble 48	48	48	2	process wastewater	X058-X003-X033-XXXX
Ruble 54	54	54	2	process wastewater	0058-0003-0013-0000
Ruble 56	56	56	2	process wastewater	0058-0003-0013-0000 X058-X003-X033-XXXX
Totals for areas that were used for application	368	368	16		
Totals for areas that were not used for application	10	10	0		
Land application area totals	378	378	16		

B. CROPS AND HARVESTS

Bernies 10

Field name: Bernies 10

09/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 09/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	153.00 *ton*	As-is		70.0	5,700.00	1,190.00	8,980.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	15.30	174.42	36.41	274.79	1,110.78

Bernies 10

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	214.00 *ton*	As-is		70.0	4,440.00	940.00	10,700.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	21.40	190.03	40.23	457.96	1,309.68

Big 68

Field name: Big 68

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 68 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	1,188.00 *ton*	As-is		70.0	4,920.00	1,200.00	9,990.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	17.47	171.91	41.93	349.06	1,111.13

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,350.00 *ton*	As-is		70.0	3,660.00	700.00	5,650.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	19.85	145.32	27.79	224.34	607.50

Long 35

Field name: Long 35

09/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 09/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	453.00 *ton*	As-is		70.0	5,310.00	1,260.00	14,400.00		11.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	12.94	137.45	32.62	372.75	893.06

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	844.00 *ton*	As-is		70.0	3,720.00	460.00	4,380.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.11	179.41	22.19	211.24	824.71

Long 57

Field name: Long 57

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	1,214.00 *ton*	As-is		70.0	3,390.00	800.00	7,390.00		9.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	21.30	144.40	34.08	314.79	1,194.83

Long 57

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,238.00 *ton*	As-is		70.0	3,690.00	760.00	6,100.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	21.72	160.29	33.01	264.98	899.18

McWharters 40

Field name: McWharters 40

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	684.00 *ton*	As-is		70.0	4,500.00	1,020.00	8,280.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	17.10	153.90	34.88	283.18	1,067.04

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	985.00 *ton*	As-is		70.0	3,960.00	450.00	5,380.00		4.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.63	195.03	22.16	264.97	664.88

Ruble 48

Field name: Ruble 48

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 48 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	1,340.00 *ton*	As-is		70.0	4,500.00	850.00	7,640.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	27.92	251.25	47.46	426.57	1,725.25

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 48 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,142.00 *ton*	As-is		70.0	3,930.00	550.00	4,600.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.79	187.00	26.17	218.88	1,113.45

Ruble 54

Field name: Ruble 54

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 54 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	1,455.00 *ton*	As-is		70.0	5,190.00	840.00	6,810.00		9.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	26.94	279.68	45.27	366.98	1,598.88

Ruble 54

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,357.00 *ton*	As-is		70.0	4,590.00	640.00	4,650.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.13	230.69	32.17	233.71	799.12

Ruble 56

Field name: Ruble 56

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 56 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	904.00 *ton*	As-is		70.0	5,390.00	1,360.00	9,280.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.50	175.00	28.00	145.25	0.00
Total actual harvest content	16.14	174.02	43.91	299.61	1,278.51

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	1,379.00 *ton*	As-is		70.0	4,200.00	540.00	4,780.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.63	206.85	26.60	235.42	1,063.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

Bernies 10 - 09/14/2022: Oats, silage-soft dough

Field name: Bernies 10

Crop: Oats, silage-soft dough

Plant date: 09/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	1,215,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irrigation Well	Ground water	1.26	0.00	0.00	420.59	360,000.00 *gal*
Application event totals		1.26	0.00	0.00	420.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	216.18	47.48	363.83	5,041.43	977,550.00 *gal*
Application event totals		216.18	47.48	363.83	5,041.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

Bernies 10 - 05/28/2023: Corn, silage

Field name: Bernies 10

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	35.15	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.03	5.79	37.93	818.48	162,925.00 *gal*
Application event totals		67.03	5.79	37.93	818.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	35.15	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.92	2.01	70.02	549.28	162,925.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	1,215,000.00 *gal*
Application event totals		71.92	2.01	70.02	575.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

Bernies 10 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.92	2.01	70.02	549.28	162,925.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	1,215,000.00 *gal*
Application event totals		71.92	2.01	70.02	575.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	810,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

Big 68 - 10/26/2022: Oats, silage-soft dough

Field name: Big 68

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Big 68 - 10/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.79	6.98	53.50	741.39	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	12.92	4,050,000.00 *gal*
Application event totals		31.79	6.98	53.50	754.31	

12/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.79	6.98	53.50	741.39	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	12.92	4,050,000.00 *gal*
Application event totals		31.79	6.98	53.50	754.31	

01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.19	4.65	35.67	494.26	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.09	5,670,000.00 *gal*
Application event totals		21.19	4.65	35.67	512.35	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.86	6.81	44.63	962.92	1,303,400.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	15.51	4,860,000.00 *gal*
Application event totals		78.86	6.81	44.63	978.43	

04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.43	3.41	22.31	481.46	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	15.51	4,860,000.00 *gal*
Application event totals		39.43	3.41	22.31	496.97	

Big 68 - 06/01/2023: Corn, silage

Big 68 - 06/01/2023: Corn, silage

Field name: Big 68

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.14	5.11	33.47	722.19	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	10.93	8,910,000.00 *gal*
Application event totals		59.14	5.11	33.47	733.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.15	0.59	20.59	161.55	325,850.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	6.96	5,670,000.00 *gal*
Application event totals		21.15	0.59	20.59	168.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.46	1.78	61.78	484.66	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	6.46	5,265,000.00 *gal*
Application event totals		63.46	1.78	61.78	491.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.31	1.18	41.19	323.11	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	14.21	4,455,000.00 *gal*
Application event totals		42.31	1.18	41.19	337.32	

Big 68 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	18.09	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	10.34	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	16.80	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.80	

Long 35 - 09/01/2022: Oats, silage-soft dough

Field name: Long 35

Crop: Oats, silage-soft dough

Plant date: 09/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	40.17	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.18	9.04	69.30	960.27	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	20.09	3,240,000.00 *gal*
Application event totals		41.18	9.04	69.30	980.36	

Long 35 - 09/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.18	9.04	69.30	960.27	651,700.00 *gal*
West Irrigation Well	Ground water	1.45	0.00	0.00	480.67	1,440,000.00 *gal*
Application event totals		42.62	9.04	69.30	1,440.94	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.60	6.62	43.35	935.41	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	15.06	2,430,000.00 *gal*
Application event totals		76.60	6.62	43.35	950.47	

Long 35 - 06/01/2023: Corn, silage

Field name: Long 35

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.45	4.96	32.51	701.56	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	25.11	4,050,000.00 *gal*
Application event totals		57.45	4.96	32.51	726.66	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	25.11	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.11	

Long 35 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.10	1.15	40.01	313.88	325,850.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	22.60	3,645,000.00 *gal*
Application event totals		41.10	1.15	40.01	336.47	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.65	1.72	60.02	470.81	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	2,835,000.00 *gal*
Application event totals		61.65	1.72	60.02	488.39	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.10	1.15	40.01	313.88	325,850.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	20.09	3,240,000.00 *gal*
Application event totals		41.10	1.15	40.01	333.96	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

Long 57 - 10/27/2022: Oats, silage-soft dough

Field name: Long 57

Crop: Oats, silage-soft dough

Plant date: 10/27/2022

Long 57 - 10/27/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.28	5.55	42.55	589.64	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	16.96	4,455,000.00 *gal*
Application event totals		25.28	5.55	42.55	606.60	

12/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.28	5.55	42.55	589.64	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	15.42	4,050,000.00 *gal*
Application event totals		25.28	5.55	42.55	605.06	

01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.28	5.55	42.55	589.64	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.50	4,860,000.00 *gal*
Application event totals		25.28	5.55	42.55	608.14	

03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.56	6.10	39.93	861.56	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	30.83	8,100,000.00 *gal*
Application event totals		70.56	6.10	39.93	892.40	

04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.02	4.06	26.61	574.14	651,433.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	12.33	3,240,000.00 *gal*
Application event totals		47.02	4.06	26.61	586.47	

Long 57 - 06/03/2023: Corn, silage

Long 57 - 06/03/2023: Corn, silage

Field name: Long 57

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.04	4.06	26.62	574.38	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	30.83	8,100,000.00 *gal*
Application event totals		47.04	4.06	26.62	605.21	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	18.50	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.50	

07/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.47	1.41	49.14	385.46	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	13.87	3,645,000.00 *gal*
Application event totals		50.47	1.41	49.14	399.34	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.47	1.41	49.14	385.46	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	12.33	3,240,000.00 *gal*
Application event totals		50.47	1.41	49.14	397.79	

Long 57 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	18.50	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.24	0.71	24.57	192.73	325,850.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.50	4,860,000.00 *gal*
Application event totals		25.24	0.71	24.57	211.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	15.42	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	15.42	

McWharters 40 - 11/06/2022: Oats, silage-soft dough

Field name: McWharters 40

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.03	5.79	37.93	818.48	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	13.18	2,430,000.00 *gal*
Application event totals		67.03	5.79	37.93	831.67	

McWharters 40 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.03	5.79	37.93	818.48	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	3,240,000.00 *gal*
Application event totals		67.03	5.79	37.93	836.06	

12/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.04	11.87	90.96	1,260.36	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	13.18	2,430,000.00 *gal*
Application event totals		54.04	11.87	90.96	1,273.54	

McWharters 40 - 06/03/2023: Corn, silage

Field name: McWharters 40

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.27	4.34	28.45	613.86	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	26.36	4,860,000.00 *gal*
Application event totals		50.27	4.34	28.45	640.23	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

McWharters 40 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.94	1.51	52.51	411.96	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.77	3,645,000.00 *gal*
Application event totals		53.94	1.51	52.51	431.73	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.94	1.51	52.51	411.96	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	3,240,000.00 *gal*
Application event totals		53.94	1.51	52.51	429.54	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.57	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.96	1.01	35.01	274.64	325,850.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.57	3,240,000.00 *gal*
Application event totals		35.96	1.01	35.01	292.22	

Ruble 48 - 10/28/2022: Oats, silage-soft dough

Field name: Ruble 48

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Ruble 48 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.02	6.59	50.53	700.20	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	4,860,000.00 *gal*
Application event totals		30.02	6.59	50.53	722.17	

01/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.02	6.59	50.53	700.20	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	14.65	3,240,000.00 *gal*
Application event totals		30.02	6.59	50.53	714.84	

02/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.01	3.30	25.27	350.10	325,850.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	25.63	5,670,000.00 *gal*
Application event totals		15.01	3.30	25.27	375.73	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.86	4.83	31.61	682.07	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.97	4,860,000.00 *gal*
Application event totals		55.86	4.83	31.61	704.04	

04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.86	4.83	31.61	682.07	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.31	4,050,000.00 *gal*
Application event totals		55.86	4.83	31.61	700.38	

Ruble 48 - 06/14/2023: Corn, silage

Ruble 48 - 06/14/2023: Corn, silage

Field name: Ruble 48

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.89	3.62	23.71	511.55	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	6,480,000.00 *gal*
Application event totals		41.89	3.62	23.71	540.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	29.29	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.95	1.26	43.76	343.30	488,775.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	31.12	6,885,000.00 *gal*
Application event totals		44.95	1.26	43.76	374.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	25.63	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.63	

Ruble 48 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.94	1.68	58.35	457.74	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.31	4,050,000.00 *gal*
Application event totals		59.94	1.68	58.35	476.04	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.94	1.68	58.35	457.74	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.31	4,050,000.00 *gal*
Application event totals		59.94	1.68	58.35	476.04	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	10.98	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.98	

Ruble 54 - 10/29/2022: Oats, silage-soft dough

Field name: Ruble 54

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.38	11.72	89.83	1,244.80	1,303,400.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	21.70	5,400,000.00 *gal*
Application event totals		53.38	11.72	89.83	1,266.49	

Ruble 54 - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.69	5.86	44.92	622.40	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	10.85	2,700,000.00 *gal*
Application event totals		26.69	5.86	44.92	633.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.69	5.86	44.92	622.40	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.36	4,320,000.00 *gal*
Application event totals		26.69	5.86	44.92	639.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.48	6.44	42.15	909.43	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	13.02	3,240,000.00 *gal*
Application event totals		74.48	6.44	42.15	922.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.65	4.29	28.10	606.28	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	13.02	3,240,000.00 *gal*
Application event totals		49.65	4.29	28.10	619.30	

Ruble 54 - 06/15/2023: Corn, silage

Field name: Ruble 54

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Ruble 54 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.65	4.29	28.10	606.28	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	26.04	6,480,000.00 *gal*
Application event totals		49.65	4.29	28.10	632.32	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	26.04	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.04	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.28	1.49	51.87	406.88	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.36	4,320,000.00 *gal*
Application event totals		53.28	1.49	51.87	424.23	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	17.36	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	17.36	

Ruble 54 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.28	1.49	51.87	406.88	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.53	4,860,000.00 *gal*
Application event totals		53.28	1.49	51.87	426.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.28	1.49	51.87	406.88	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	19.53	4,860,000.00 *gal*
Application event totals		53.28	1.49	51.87	426.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	10.85	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.85	

Ruble 56 - 10/28/2022: Oats, silage-soft dough

Field name: Ruble 56

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.47	11.30	86.63	1,200.34	1,303,400.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	20.92	5,400,000.00 *gal*
Application event totals		51.47	11.30	86.63	1,221.26	

Ruble 56 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.74	5.65	43.31	600.17	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	10.46	2,700,000.00 *gal*
Application event totals		25.74	5.65	43.31	610.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	25.74	5.65	43.31	600.17	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	16.74	4,320,000.00 *gal*
Application event totals		25.74	5.65	43.31	616.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.82	6.21	40.64	876.95	977,550.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	12.55	3,240,000.00 *gal*
Application event totals		71.82	6.21	40.64	889.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.88	4.14	27.10	584.63	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	12.55	3,240,000.00 *gal*
Application event totals		47.88	4.14	27.10	597.19	

Ruble 56 - 06/16/2023: Corn, silage

Field name: Ruble 56

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Ruble 56 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.88	4.14	27.10	584.63	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	23.01	5,940,000.00 *gal*
Application event totals		47.88	4.14	27.10	607.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	16.74	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
NH3	Solid commercial fertilizer	49.20	0.00	0.00	0.00	
Application event totals		49.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.37	1.44	50.01	392.34	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	17.78	4,590,000.00 *gal*
Application event totals		51.37	1.44	50.01	410.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	16.74	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	16.74	

Ruble 56 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.37	1.44	50.01	392.34	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.83	4,860,000.00 *gal*
Application event totals		51.37	1.44	50.01	411.17	

09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.37	1.44	50.01	392.34	651,700.00 *gal*
TID Lateral 4	Ground water	0.00	0.00	0.00	18.83	4,860,000.00 *gal*
Application event totals		51.37	1.44	50.01	411.17	

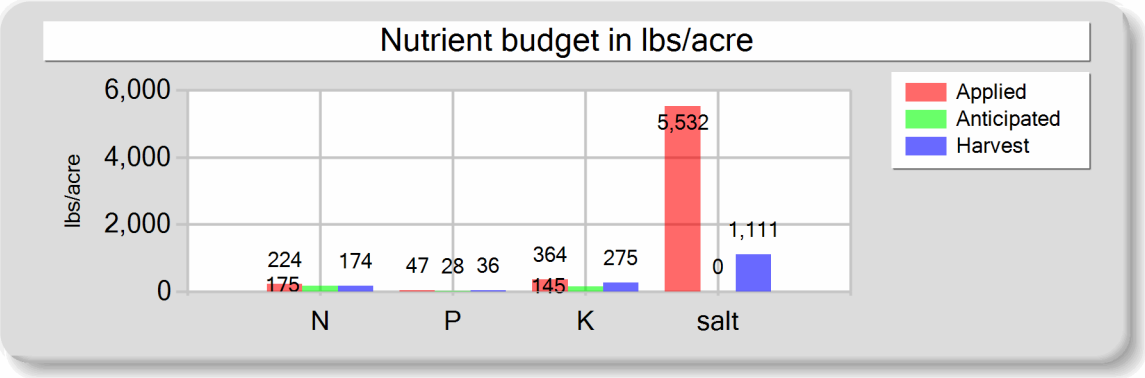
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 4	Ground water	0.00	0.00	0.00	10.46	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	10.46	

B. NUTRIENT BUDGET

Bernies 10 - 09/14/2022: Oats, silage-soft dough

Field name: Bernies 10 Crop: Oats, silage-soft dough Plant date: 09/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.18	47.48	363.83	5,041.43
Fresh water	1.26	0.00	0.00	490.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.44	47.48	363.83	5,532.32
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	174.42	36.41	274.79	1,110.78
Nutrient balance	50.02	11.06	89.04	4,421.54
Applied to removed ratio	1.29	1.30	1.32	4.98

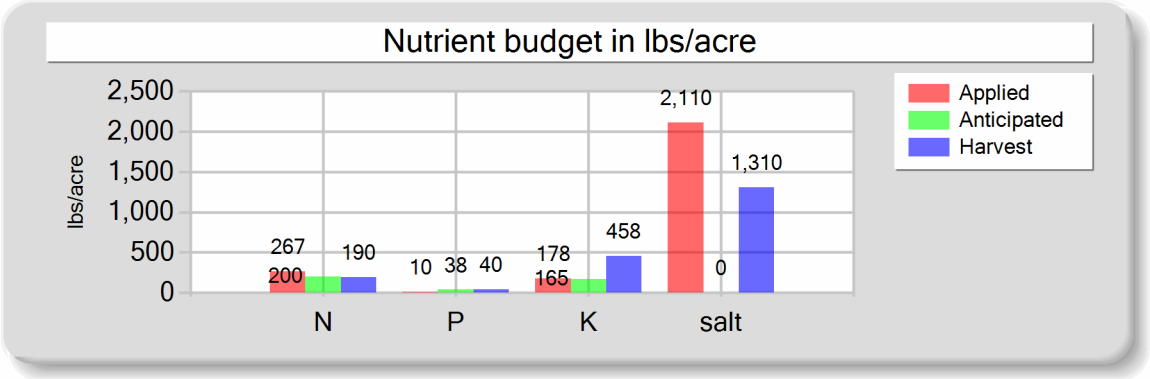
Fresh water applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
13.26	*inches/acre*

Process wastewater applied	
977,550.00	*gallons*
36.00	*acre-inches*
3.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bernies 10 - 05/28/2023: Corn, silage

Field name: Bernies 10 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.88	9.82	177.97	1,917.05
Fresh water	0.00	0.00	0.00	193.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.08	9.82	177.97	2,110.37
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	190.03	40.23	457.96	1,309.68
Nutrient balance	77.04	-30.42	-279.99	800.69
Applied to removed ratio	1.41	0.24	0.39	1.61

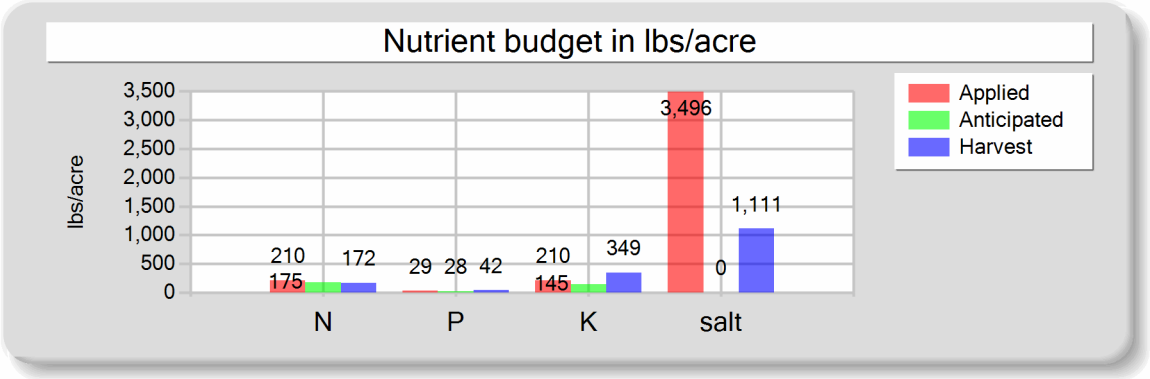
Fresh water applied	
8,910,000.00	*gallons*
328.13	*acre-inches*
32.81	*inches/acre*

Process wastewater applied	
488,775.00	*gallons*
18.00	*acre-inches*
1.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big 68 - 10/26/2022: Oats, silage-soft dough

Field name: Big 68 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	203.06	28.84	209.62	3,421.42
Fresh water	0.00	0.00	0.00	74.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.06	28.84	209.62	3,496.37
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	171.91	41.93	349.06	1,111.13
Nutrient balance	38.15	-13.09	-139.44	2,385.24
Applied to removed ratio	1.22	0.69	0.60	3.15

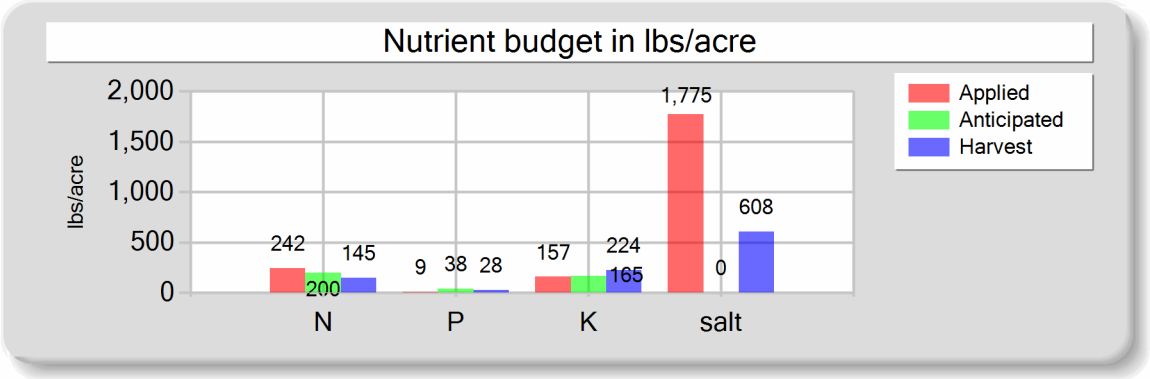
Fresh water applied	
23,490,000.00	*gallons*
865.06	*acre-inches*
12.72	*inches/acre*

Process wastewater applied	
4,561,900.00	*gallons*
168.00	*acre-inches*
2.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big 68 - 06/01/2023: Corn, silage

Field name: Big 68 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.07	8.66	157.03	1,691.51
Fresh water	0.00	0.00	0.00	83.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.27	8.66	157.03	1,775.31
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	145.32	27.79	224.34	607.50
Nutrient balance	96.94	-19.13	-67.30	1,167.81
Applied to removed ratio	1.67	0.31	0.70	2.92

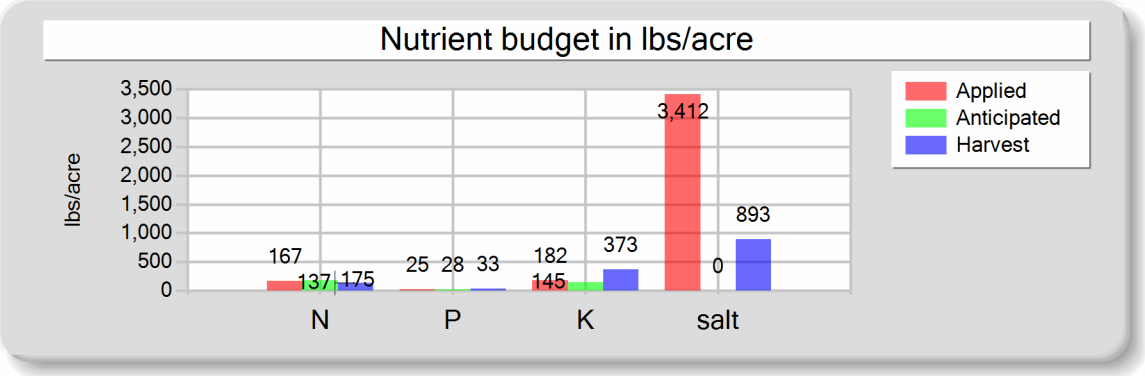
Fresh water applied	
38,475,000.00	*gallons*
1,416.90	*acre-inches*
20.84	*inches/acre*

Process wastewater applied	
2,932,650.00	*gallons*
108.00	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Long 35 - 09/01/2022: Oats, silage-soft dough

Field name: Long 35 Crop: Oats, silage-soft dough Plant date: 09/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	158.96	24.71	181.95	2,855.96
Fresh water	1.45	0.00	0.00	555.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	167.40	24.71	181.95	3,411.95
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	137.45	32.62	372.75	893.06
Nutrient balance	29.95	-7.91	-190.80	2,518.89
Applied to removed ratio	1.22	0.76	0.49	3.82

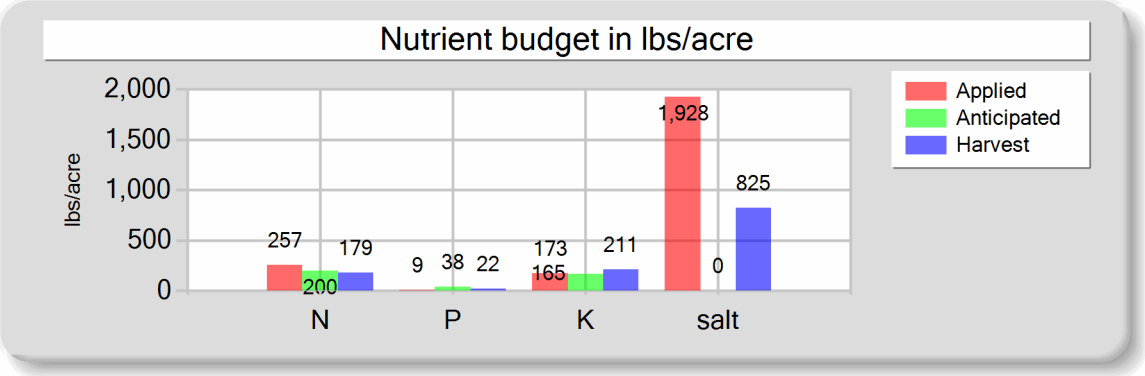
Fresh water applied	
13,590,000.00	*gallons*
500.47	*acre-inches*
14.30	*inches/acre*

Process wastewater applied	
1,955,100.00	*gallons*
72.00	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Long 35 - 06/01/2023: Corn, silage

Field name: Long 35 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.30	8.99	172.55	1,800.12
Fresh water	0.00	0.00	0.00	128.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.50	8.99	172.55	1,928.17
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	179.41	22.19	211.24	824.71
Nutrient balance	78.09	-13.20	-38.69	1,103.46
Applied to removed ratio	1.44	0.41	0.82	2.34

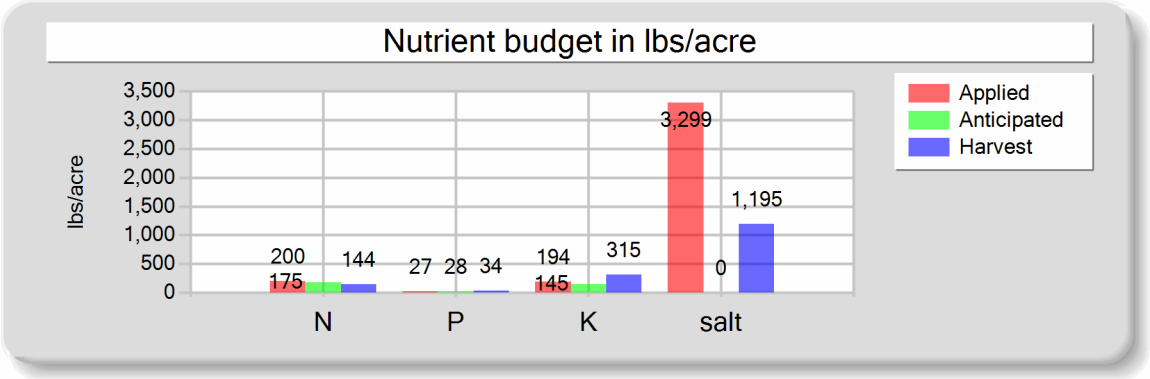
Fresh water applied	
20,655,000.00	*gallons*
760.65	*acre-inches*
21.73	*inches/acre*

Process wastewater applied	
1,629,250.00	*gallons*
60.00	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Long 57 - 10/27/2022: Oats, silage-soft dough

Field name: Long 57 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.43	26.82	194.20	3,204.63
Fresh water	0.00	0.00	0.00	94.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.43	26.82	194.20	3,298.67
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	144.40	34.08	314.79	1,194.83
Nutrient balance	56.02	-7.26	-120.59	2,103.83
Applied to removed ratio	1.39	0.79	0.62	2.76

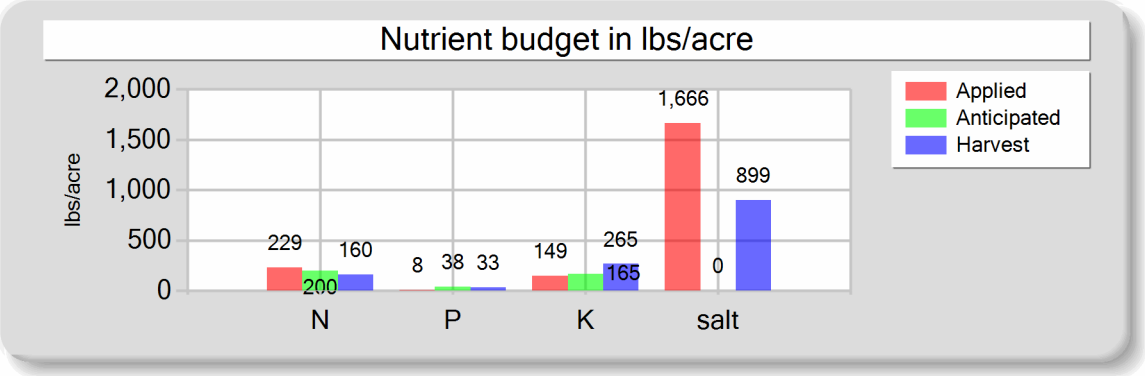
Fresh water applied	
24,705,000.00	*gallons*
909.80	*acre-inches*
15.96	*inches/acre*

Process wastewater applied	
3,584,083.00	*gallons*
131.99	*acre-inches*
2.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

Long 57 - 06/03/2023: Corn, silage

Field name: Long 57 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.22	7.59	149.46	1,538.03
Fresh water	0.00	0.00	0.00	127.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.42	7.59	149.46	1,665.98
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	160.29	33.01	264.98	899.18
Nutrient balance	69.13	-25.42	-115.51	766.80
Applied to removed ratio	1.43	0.23	0.56	1.85

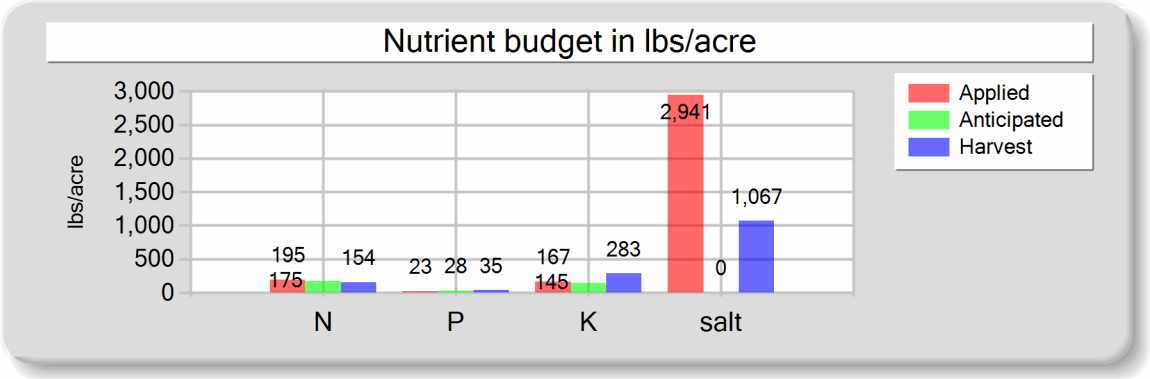
Fresh water applied	
33,615,000.00	*gallons*
1,237.93	*acre-inches*
21.72	*inches/acre*

Process wastewater applied	
2,280,950.00	*gallons*
84.00	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

McWharters 40 - 11/06/2022: Oats, silage-soft dough

Field name: McWharters 40 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	188.10	23.45	166.82	2,897.33
Fresh water	0.00	0.00	0.00	43.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.10	23.45	166.82	2,941.26
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	153.90	34.88	283.18	1,067.04
Nutrient balance	41.20	-11.43	-116.35	1,874.22
Applied to removed ratio	1.27	0.67	0.59	2.76

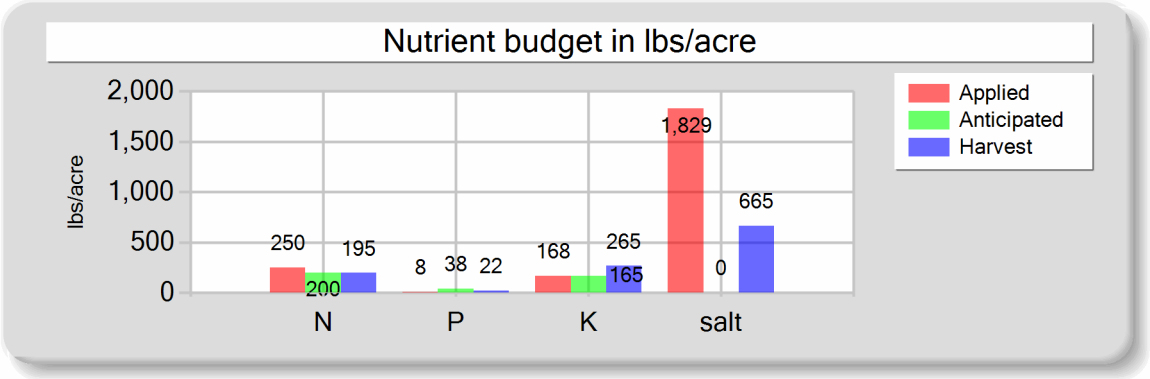
Fresh water applied	
8,100,000.00	*gallons*
298.30	*acre-inches*
7.46	*inches/acre*

Process wastewater applied	
2,280,950.00	*gallons*
84.00	*acre-inches*
2.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

McWharters 40 - 06/03/2023: Corn, silage

Field name: McWharters 40 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.12	8.37	168.49	1,712.43
Fresh water	0.00	0.00	0.00	116.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.32	8.37	168.49	1,828.86
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	195.03	22.16	264.97	664.88
Nutrient balance	55.29	-13.79	-96.48	1,163.98
Applied to removed ratio	1.28	0.38	0.64	2.75

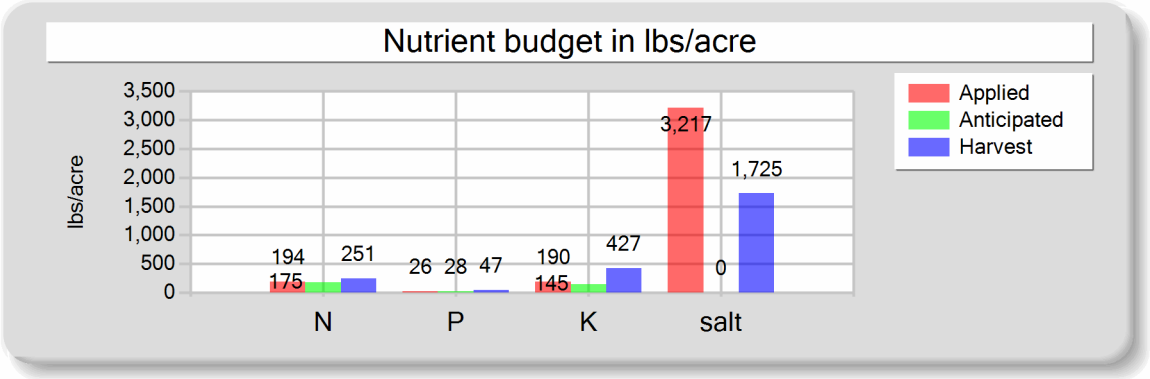
Fresh water applied	
21,465,000.00	*gallons*
790.48	*acre-inches*
19.76	*inches/acre*

Process wastewater applied	
1,792,175.00	*gallons*
66.00	*acre-inches*
1.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ruble 48 - 10/28/2022: Oats, silage-soft dough

Field name: Ruble 48 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.78	26.14	189.55	3,114.64
Fresh water	0.00	0.00	0.00	102.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.78	26.14	189.55	3,217.16
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	251.25	47.46	426.57	1,725.25
Nutrient balance	-57.47	-21.32	-237.01	1,491.91
Applied to removed ratio	0.77	0.55	0.44	1.86

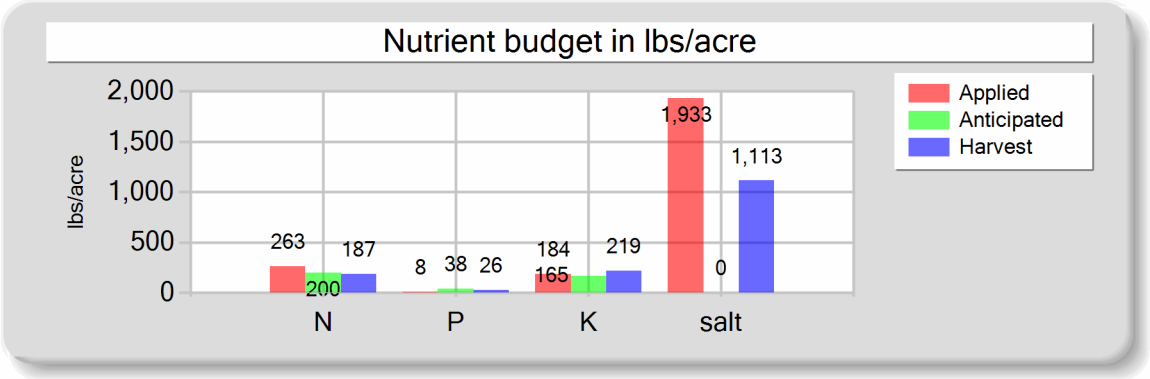
Fresh water applied	
22,680,000.00	*gallons*
835.23	*acre-inches*
17.40	*inches/acre*

Process wastewater applied	
2,932,650.00	*gallons*
108.00	*acre-inches*
2.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ruble 48 - 06/14/2023: Corn, silage

Field name: Ruble 48 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.72	8.23	184.17	1,770.32
Fresh water	0.00	0.00	0.00	162.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.92	8.23	184.17	1,933.26
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	187.00	26.17	218.88	1,113.45
Nutrient balance	75.91	-17.94	-34.71	819.81
Applied to removed ratio	1.41	0.31	0.84	1.74

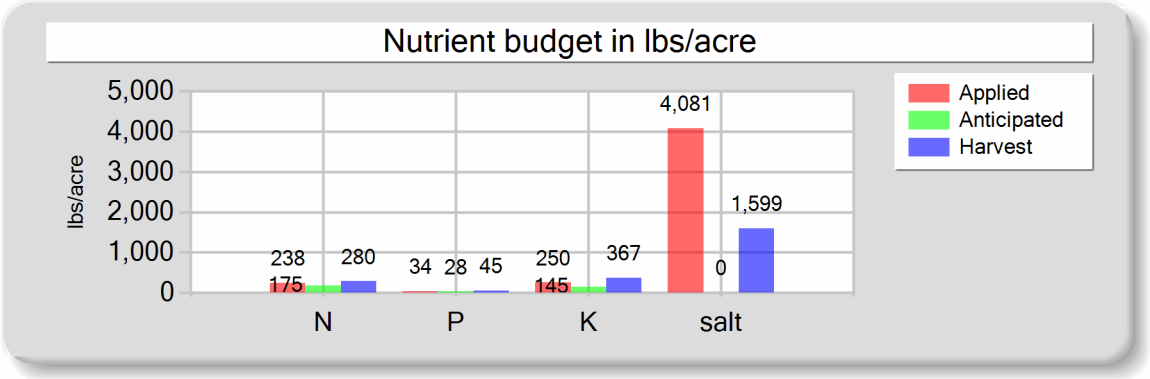
Fresh water applied	
36,045,000.00	*gallons*
1,327.41	*acre-inches*
27.65	*inches/acre*

Process wastewater applied	
2,280,950.00	*gallons*
84.00	*acre-inches*
1.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ruble 54 - 10/29/2022: Oats, silage-soft dough

Field name: Ruble 54 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.88	34.17	249.92	4,005.31
Fresh water	0.00	0.00	0.00	75.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	237.88	34.17	249.92	4,081.25
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	279.68	45.27	366.98	1,598.88
Nutrient balance	-41.80	-11.10	-117.07	2,482.36
Applied to removed ratio	0.85	0.75	0.68	2.55

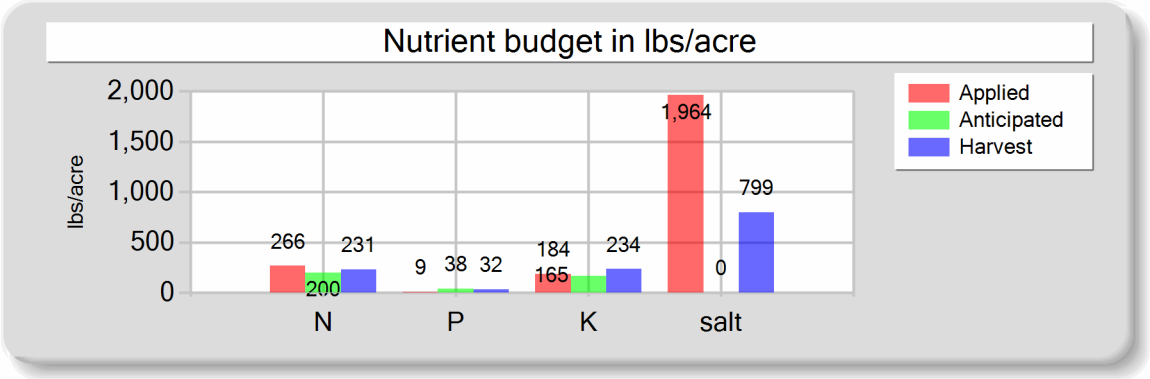
Fresh water applied	
18,900,000.00	*gallons*
696.02	*acre-inches*
12.89	*inches/acre*

Process wastewater applied	
4,236,050.00	*gallons*
156.00	*acre-inches*
2.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ruble 54 - 06/15/2023: Corn, silage

Field name: Ruble 54 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	209.48	8.76	183.70	1,826.91
Fresh water	0.00	0.00	0.00	136.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.68	8.76	183.70	1,963.60
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	230.69	32.17	233.71	799.12
Nutrient balance	34.99	-23.40	-50.01	1,164.48
Applied to removed ratio	1.15	0.27	0.79	2.46

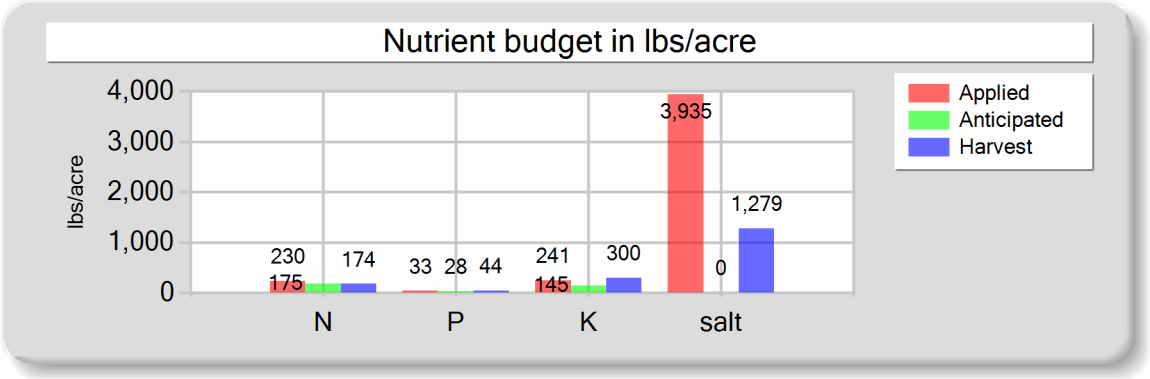
Fresh water applied	
34,020,000.00	*gallons*
1,252.84	*acre-inches*
23.20	*inches/acre*

Process wastewater applied	
2,606,800.00	*gallons*
96.00	*acre-inches*
1.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ruble 56 - 10/28/2022: Oats, silage-soft dough

Field name: Ruble 56 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.64	32.95	240.99	3,862.26
Fresh water	0.00	0.00	0.00	73.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.64	32.95	240.99	3,935.49
Anticipated crop nutrient removal	175.00	28.00	145.25	0.00
Actual crop nutrient removal	174.02	43.91	299.61	1,278.51
Nutrient balance	55.62	-10.96	-58.62	2,656.97
Applied to removed ratio	1.32	0.75	0.80	3.08

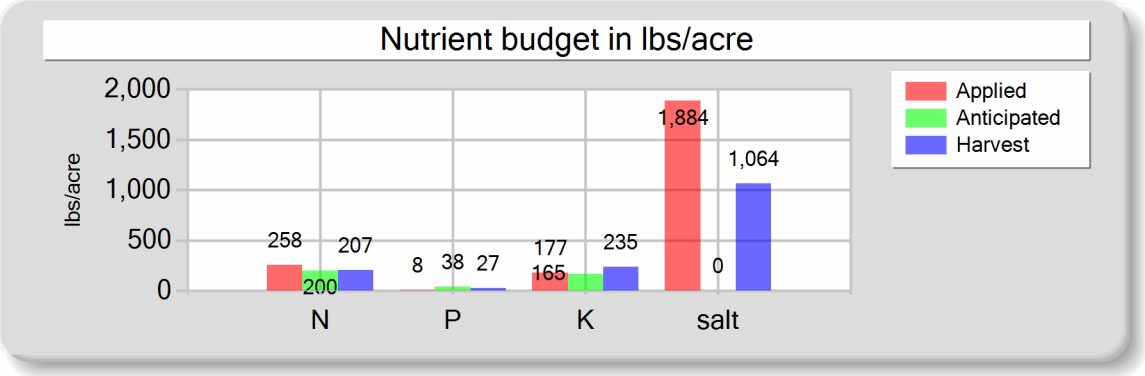
Fresh water applied	
18,900,000.00	*gallons*
696.02	*acre-inches*
12.43	*inches/acre*

Process wastewater applied	
4,236,050.00	*gallons*
156.00	*acre-inches*
2.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

Ruble 56 - 06/16/2023: Corn, silage

Field name: Ruble 56 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	49.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	202.00	8.45	177.14	1,761.66
Fresh water	0.00	0.00	0.00	122.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.20	8.45	177.14	1,884.06
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	206.85	26.60	235.42	1,063.80
Nutrient balance	51.35	-18.15	-58.28	820.26
Applied to removed ratio	1.25	0.32	0.75	1.77

Fresh water applied	
31,590,000.00	*gallons*
1,163.35	*acre-inches*
20.77	*inches/acre*

Process wastewater applied	
2,606,800.00	*gallons*
96.00	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 11/04/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	7,320.00	45,000.00							42.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/04/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	265.00	180.00	0.00	0.00	58.20	446.00								5.80	6,180
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	493.00	171.00	0.00	0.00	42.60	279.00								4,450.00	6,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 07/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	529.00	227.00	0.00	0.00	14.80	515.00								6,410.00	4,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 10/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	230.00	148.00	0.00	0.00	32.20	290.00								4.93	3,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

East Irrigation Well

Fresh Water

Sample description: Fresh Water

Sample date: 12/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	25.00	0.00	25.00								200.00	1,140
DL	0.50	0.50	0.50								1.00	10

TID Lateral 4

TID Lateral 4

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.00	26
DL	0.50	0.50	0.50								1.00	10

West Irrigation Well

Fresh Water

Sample description: Fresh Water

Sample date: 12/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.21	0.00	4.21								2,680.00	1,400
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Bernies 10 - 09/14/2022: Oats, silage-soft dough

Bernies 10 - 09/14/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	1,190.00	8,980.00		12.10
DL	100.00	100.00	100.00		0.01

Bernies 10 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,440.00	940.00	10,700.00		10.20
DL	100.00	100.00	100.00		0.01

Big 68 - 10/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,920.00	1,200.00	9,990.00		10.60
DL	100.00	100.00	100.00		0.01

Big 68 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,660.00	700.00	5,650.00		5.10
DL	100.00	100.00	100.00		0.01

Long 35 - 09/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,310.00	1,260.00	14,400.00		11.50
DL	100.00	100.00	100.00		0.01

Long 35 - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,720.00	460.00	4,380.00		5.70
DL	100.00	100.00	100.00		0.01

Long 57 - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	800.00	7,390.00		9.35
DL	100.00	100.00	100.00		0.01

Long 57 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,690.00	760.00	6,100.00		6.90
DL	100.00	100.00	100.00		0.01

McWharters 40 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	450.00	5,380.00		4.50
DL	100.00	100.00	100.00		0.01

Ruble 48 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	850.00	7,640.00		3.09
DL	100.00	100.00	100.00		0.01

Ruble 48 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,930.00	550.00	4,600.00		7.80
DL	100.00	100.00	100.00		0.01

Ruble 54 - 10/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,190.00	840.00	6,810.00		9.89
DL	100.00	100.00	100.00		0.01

Ruble 54 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,590.00	640.00	4,650.00		5.30
DL	100.00	100.00	100.00		0.01

Ruble 56 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,390.00	1,360.00	9,280.00		13.20
DL	100.00	100.00	100.00		0.01

Ruble 56 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	540.00	4,780.00		7.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

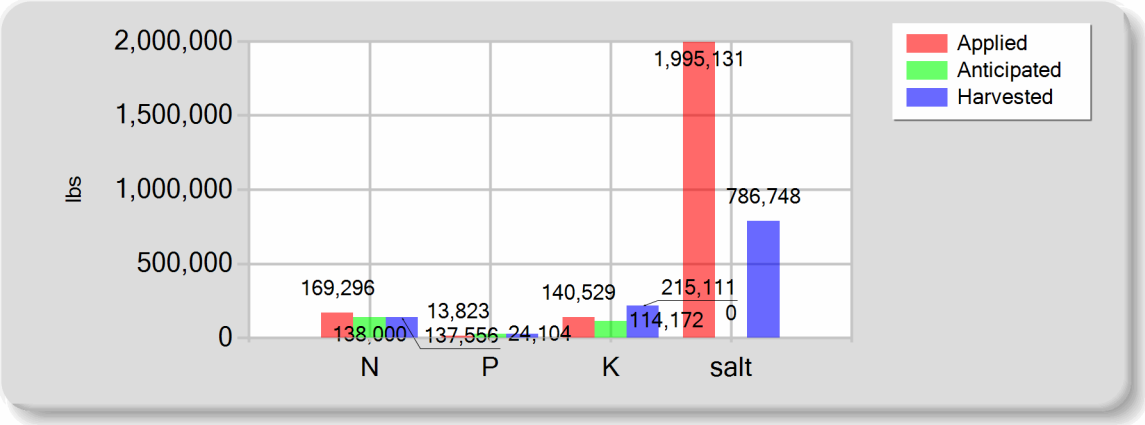
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

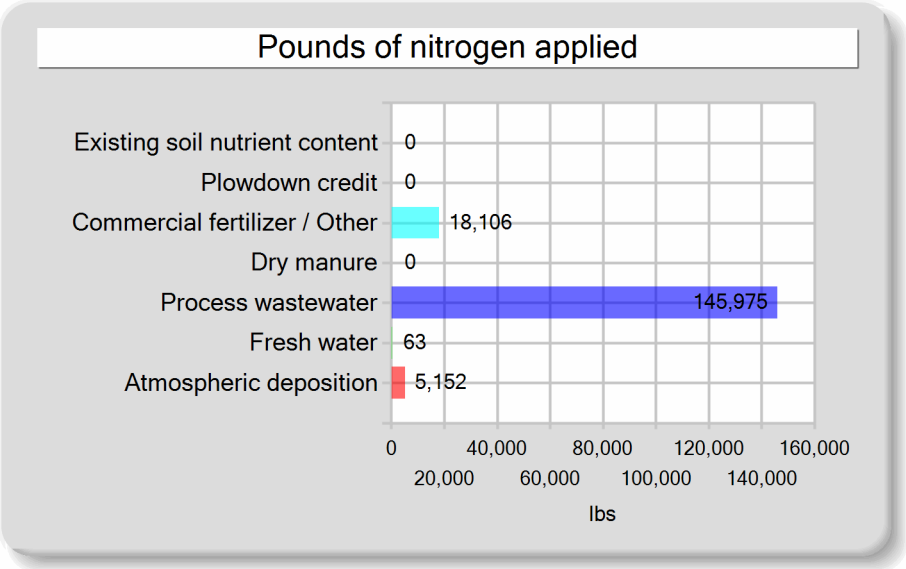
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

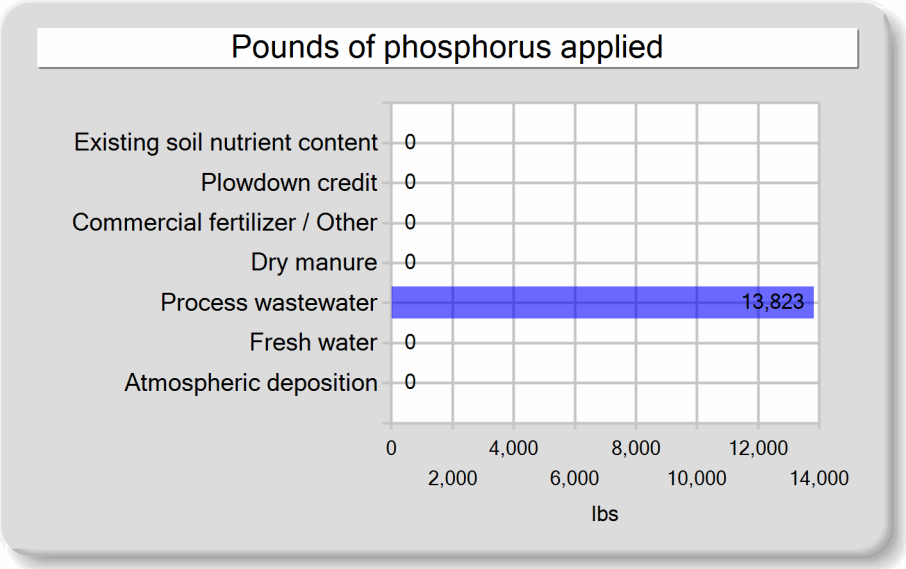
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18,105.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	145,974.69	13,823.32	140,528.58	1,899,306.15
Fresh water	63.24	0.00	0.00	95,824.97
Atmospheric deposition	5,152.00	0.00	0.00	0.00
Total nutrients applied	169,295.53	13,823.32	140,528.58	1,995,131.12
Anticipated crop nutrient removal	138,000.00	24,104.00	114,172.00	0.00
Actual crop nutrient removal	137,556.22	25,195.48	215,111.00	786,747.90
Nutrient balance	31,739.31	-11,372.16	-74,582.42	1,208,383.22
Applied to removed ratio	1.23	0.55	0.65	2.54

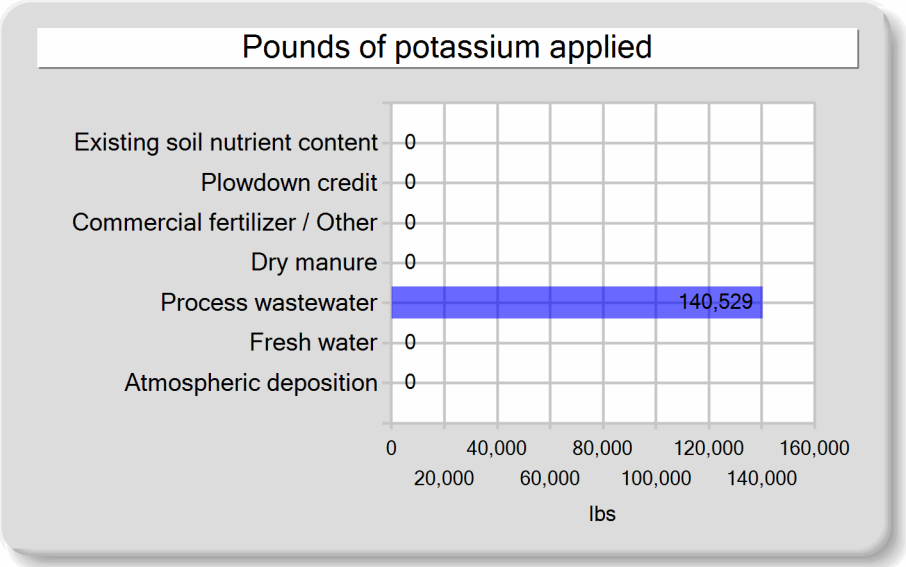
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

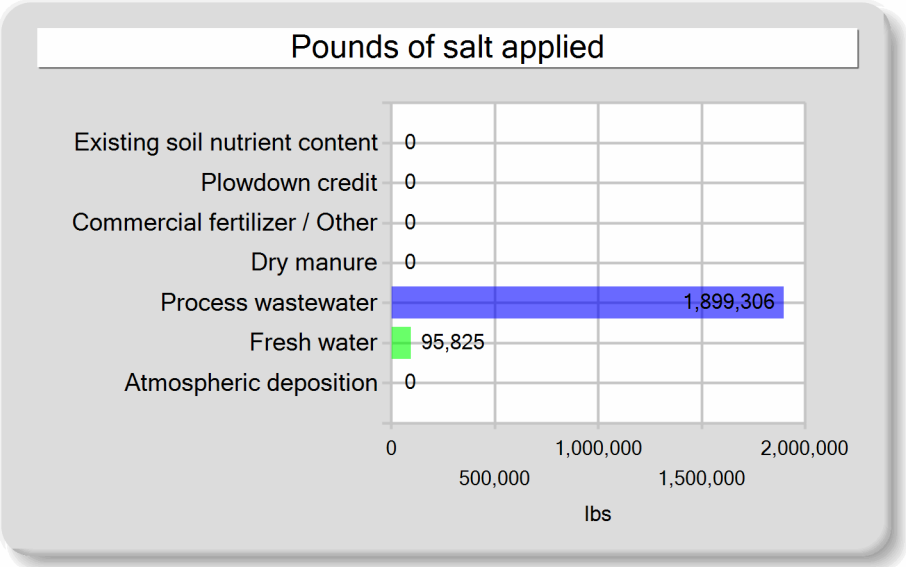


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Poor corn yield on Big 68 field in 2023 resulted in a nitrogen application to removal ratio exceeding 1.6.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Albert Mendes (Apr 19, 2024 04:32 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Albert Mendes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 19, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Albert Mendes

Name of Dairy Facility: Albert Mendes Dairy

Facility Address:

1100 Ruble RD	Crows Landing	Stanislaus	95313
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Albert Mendes (120) 967-8027
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: 5 Star Custom Spreading

Address of Hauling Company/Person:

219 Jensen Rd	Gustine	CA	95322
Number and Street	City	State	Zip Code

Contact Person: Eric Rocha (209) 854-6251
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

W Main Ranch (209) 000-0000
Name Phone Number

1512 W Main Street	Crows Landing	CA	95313
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

X058-X003-X018-XXXX Stanislaus
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 04/29/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 432.00 tons

Manure Solids Content: 40.4 %

Method used to determine amount of manure:

Scale Weights multiplied by Load Counts

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Albert Mendes (Apr 19, 2024 04:32 PDT)

Apr 19, 2024

Operator Signature | Date

Eric Rocha

Hauler Signature | Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Albert Mendes

Name of Dairy Facility: Albert Mendes Dairy

Facility Address:

1100 Ruble RD	Crows Landing	Stanislaus	95313
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Albert Mendes (120) 967-8027
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: 5 Star Custom Spreading

Address of Hauling Company/Person:

219 Jensen Rd	Gustine	CA	95322
Number and Street	City	State	Zip Code

Contact Person: Eric Rocha (209) 854-6251
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

RP Feedlot (209) 000-0000
Name Phone Number

12467 Carpenter Rd	Crows Landing	CA	95313
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

12467 Carpenter RD	Crows Landing	95313
Address	City	Zip Code

Stanislaus
Street and nearest cross street (if no address) County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 06/27/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,070.00 tons

Manure Solids Content: 40.4 %

Method used to determine amount of manure:

Scale Weight multiplied by loads

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Albert Mendes (Apr 19, 2024 04:32 PDT)

Apr 19, 2024

Operator Signature | Date

Eric Rocha

Hauler Signature | Date

Denele Analytical, Inc.

1232 South Ave. Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

Compliance Analysis Report

Customer
Albert Mendes Dairy
1100 Ruble Rd.
Crowslanding, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
East Well	T0356010A	Electrical Conductivity (EC)	EPA 120.1 and	2.31	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	24.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1480	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Nitrate Nitrogen (NO3-N)				

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ELAP Certificate No.: 2714

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
West Well	T0356010B	Electrical Conductivity (EC)	EPA 120.1 and	2.32	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	21.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1480	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Nitrate Nitrogen (NO3-N)				

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Grower

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Simmons Well	T0356010C	Electrical Conductivity (EC)	EPA 120.1 and	2.35	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.43	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1500	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Nitrate Nitrogen (NO3-N)				

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Compliance Analysis Report

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Albert Mendes Dairy
1100 Ruble Rd.
Crowslanding, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
West Irr. Well	T0356010D	Electrical Conductivity (EC)	EPA 120.1 and	2.68	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	0.000	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.21	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1400	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	4.21	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1720	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Nitrate Nitrogen (NO3-N)						
Total Dissolved Solids (TDS)	100			165 (T0356013G)	170 (Dup)	2.99
Total Nitrogen (TN)	109	109 (T0356005A)	109 (T0356005A)			0.000

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Compliance Analysis Report

Customer
Albert Mendes Dairy
1100 Ruble Rd.
Crowslanding, CA 95313

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 16.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
East Irr. Well	T0356010E	Electrical Conductivity (EC)	EPA 120.1 and	2.00	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	0.000	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	25.0	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1140	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	25.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1280	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Nitrate Nitrogen (NO3-N)						
Total Dissolved Solids (TDS)	100			165 (T0356013G)	170 (Dup)	2.99
Total Nitrogen (TN)	109	109 (T0356005A)	109 (T0356005A)			0.000

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Jesus Well	T0356010F	Electrical Conductivity (EC)	EPA 120.1 and	1.40	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	896	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Nitrate Nitrogen (NO3-N)				

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Grower

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/18/23 2:05 PM
SUBMITTED BY:
ANALYZED DATE: 12/22/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Francisco Well	T0356010G	Electrical Conductivity (EC)	EPA 120.1 and	2.77	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 C-2011	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1770	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	105	102 (null)	102 (null)	0.000
Nitrate Nitrogen (NO3-N)				

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Albert Mendes Dairy
Mailing address: 1100 Ruble Rd
(not facility address)
Phone#:

Authorized Copy Released To: F&R Ag
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- ~~H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS~~
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- ~~M1 % Moisture~~
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12-17	10:40	East Well		16.4°		X				
2		10:43	West Well		16.4°		X				
3		11:00	Simmons Well		16.2°		X				
4		11:05	West Irr Well		16.4°						
5		11:20	East Irr Well		16.4°						H7
6		11:43	Jesus Well		16.2°		X				H7
7		11:25	Francisco Well		16.2°		X				
8		11:15	Lagoon		16.6°						
9		11:45	Solid Manure								H8
10											M7

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	[illegible]			
2nd	[illegible]	DENELE ANALYTICAL, INC.	12/19/22 2:05	

(LABORATORY USE ONLY)

Logged In By:	LABORATORY NUMBER
	1035600[redacted]10

AMD 2023 Annual Report Complete

Final Audit Report 2024-04-19

Created:	2024-04-17
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAATn5LDlVEq7O6KTKUaJixRWVuNUN6rO7-

"AMD 2023 Annual Report Complete" History

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Signer bertm55@msn.com entered name at signing as Albert Mendes
2024-04-19 - 11:32:33 AM GMT

Document e-signed by Albert Mendes (bertm55@msn.com)
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